But if you do so, neither rice nor wheat flour will fit in the crate, and their combined calories count exceeds, by far, any other combination you can create within the remaining 200 kgs left. The best value you get with this strategy, is **1,749,800 calories**, picking potatoes, strawberry jam and peanut butter.

So, what would have looked like the most natural approach, a *greedy*[6] algorithm that at each step chooses the best immediate option, does not work. This problem needs to be carefully thought through.

Time for brainstorming. You gather your logistic team and together look for a solution.

Soon enough, someone suggests that, rather than the total calories, you should look at the calories per kg. So, you update table 1.1 with a new column, and sort it accordingly: the result is shown in table 1.2.

Table 1.2 The list in table 1.1, sorted by calories per kg.

Food	Weight (kgs)	Total calories	Calories per kg
Peanut butter	20	117,800	5,890
Rice	300	1,122,000	3,740
Wheat Flour	400	1,444,000	3,610
Strawberry jam	50	130,000	2,600
Beans (can)	300	690,000	2,300
Potatoes	800	1,501,600	1,877